Agent_5

https://gemini.google.com/app/985bedf78315a99b

User prompt: you are part of a big mission and your task is given in the md file

Response: 1. Vertical Normalization Integral Evaluation

Starting from the vertical wavefunction component $Y(y) = C_y \mathrm{Ai}(z(y))$, the dimensionless position coordinate $z(y)$ is defined as:

$$z(y) = \frac{y - \frac{E_y}{mg}}{l_g} = \left(\frac{2m^2 g}{\hbar^2}\right)^{1/3} y - \frac{E_y}{m g l_g}$$

where the gravitational characteristic length scale l_g is:

$$l_g = \left(\frac{\hbar^2}{2m^2 g}\right)^{1/3}$$

The classical vertical energy component at launch is $E_y = \frac{1}{2} m u^2 \sin^2\theta$. Evaluated at ground level ($y = 0$), $z(0) \equiv z_0$:

$$z_0 = -\frac{E_y}{m g l_g} = -\frac{\frac{1}{2} m u^2 \sin^2\theta}{mg \left(\frac{\hbar^2}{2m^2 g}\right)^{1/3}} = -\left(\frac{m u^3 \sin^3\theta}{2g\hbar}\right)^{2/3}$$

Applying the differential change of variables $dy = l_g \, dz$, the vertical normalization integral transforms as:

$$\int_0^\infty |Y(y)|^2 \, dy = 1 \implies C_y^2 l_g \int_{z_0}^\infty \mathrm{Ai}^2(z) \, dz = 1$$

2. Explicit Calculation of Normalization Constant C_y

Using the indefinite integral identity for Airy functions:

$$\int \mathrm{Ai}^2(z) \, dz = z\mathrm{Ai}^2(z) - \mathrm{Ai}'^2(z)$$

Evaluating over the interval $[z_0, \infty)$:

$$\int_{z_0}^\infty \mathrm{Ai}^2(z) \, dz = \left[z\mathrm{Ai}^2(z) - \mathrm{Ai}'^2(z)\right]_{z_0}^\infty$$

Because $\mathrm{Ai}(z) \to 0$ and $\mathrm{Ai}'(z) \to 0$ exponentially as $z \to \infty$, the upper limit vanishes, yielding:

$$\int_{z_0}^\infty \mathrm{Ai}^2(z) \, dz = 0 - \left(z_0\mathrm{Ai}^2(z_0) - \mathrm{Ai}'^2(z_0)\right) = \mathrm{Ai}'^2(z_0) - z_0\mathrm{Ai}^2(z_0)$$

Substituting this result back into the normalization condition gives:

$$C_y^2 l_g \left[\mathrm{Ai}'^2(z_0) - z_0\mathrm{Ai}^2(z_0)\right] = 1$$

Solving explicitly for C_y:

$$C_y = \frac{1}{l_g \left[\mathrm{Ai}'^2(z_0) - z_0\mathrm{Ai}^2(z_0)\right]} = \frac{\left(\frac{2m^2 g}{\hbar^2}\right)^{1/6}}{\mathrm{Ai}'^2(z_0) - z_0\mathrm{Ai}^2(z_0)}$$

Hard Boundary Condition Note: If the quantum state strictly satisfies the infinite potential wall boundary condition at $y = 0$ ($Y(0) = 0 \implies \mathrm{Ai}(z_0) = 0$), z_0 is a zero of the Airy function ($z_0 = a_n$). The expression simplifies to:

$$C_y = \frac{1}{l_g |\mathrm{Ai}'(z_0)|} = \frac{\left(\frac{2m^2 g}{\hbar^2}\right)^{1/6}}{|\mathrm{Ai}'(z_0)|}$$

3. Synthesis of 2D Wavefunction $\psi(x, y)$

The horizontal potential $V(x) = 0$ corresponds to a free particle moving along x with momentum $p_x = mu\cos\theta$ and wavenumber $k_x = \frac{mu\cos\theta}{\hbar}$:

$$X(x) = C_x \exp\left(i \frac{mu\cos\theta}{\hbar} x\right)$$

where C_x is the horizontal normalization factor (e.g., $C_x = \frac{1}{L}$ for spatial box normalization of length L, or $C_x = \frac{1}{2\pi\hbar}$ for continuous momentum delta-normalization).

Combining $X(x)$ and $Y(y)$:

$$\psi(x,y) = X(x)Y(y) = C_x C_y \exp\left(i\frac{mu\cos\theta}{\hbar}x\right)\mathrm{Ai}(z(y))$$

4. Fully Substituted Unified Equations

Substituting $z(y)$, z_0, and C_y directly in terms of the fundamental parameters $(m, u, \theta, g, \hbar)$:

General Normalized Wavefunction:

$$\psi(x,y) = C_x \frac{\left(\frac{2m^2 g}{\hbar^2}\right)^{1/6}}{\mathrm{Ai}'^2\left(-\left(\frac{mu^3\sin^3\theta}{2g\hbar}\right)^{2/3}\right) + \left(\frac{mu^3\sin^3\theta}{2g\hbar}\right)^{2/3}\mathrm{Ai}^2\left(-\left(\frac{mu^3\sin^3\theta}{2g\hbar}\right)^{2/3}\right)} \exp\left(i\frac{mu\cos\theta}{\hbar}x\right)\mathrm{Ai}\left(\left(\frac{2m^2 g}{\hbar^2}\right)^{1/3} y - \left(\frac{mu^3\sin^3\theta}{2g\hbar}\right)^{2/3}\right)$$

Quantized Boundary-Enforced Wavefunction ($\mathrm{Ai}(z_0) = 0$):

$$\psi(x,y) = \frac{C_x \left(\frac{2m^2 g}{\hbar^2}\right)^{1/6}}{\left|\mathrm{Ai}'\left(-\left(\frac{mu^3\sin^3\theta}{2g\hbar}\right)^{2/3}\right)\right|} \exp\left(i\frac{mu\cos\theta}{\hbar}x\right)\mathrm{Ai}\left(\left(\frac{2m^2 g}{\hbar^2}\right)^{1/3} y - \left(\frac{mu^3\sin^3\theta}{2g\hbar}\right)^{2/3}\right)$$
